

Zechen LI

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Education

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| Ph.D. University of New South Wales , Sydney, Australia
Computer Science Supervisor: Prof. Flora Salim, Dr. Hao Xue | 2023 – 2027 (expected) |
| M.S. Northeastern University , Boston, USA
Data Analytics Engineering GPA: 3.917/4.0 | 2019 – 2021 |
| B.Eng. Southern Medical University , Guangzhou, China
Biomedical Engineering Minor: Information Technology GPA: 3.09/4.0 | 2015 – 2019 |

Research Interests

- Multimodal learning, wearable sensor–language alignment, and LLMs for health and human activity recognition.
- Advancing verifiable and structured reasoning on wearable health data through domain knowledge integration and evidence retrieval with LLMs.

Selected Publications

(*: equal contribution)

- Hada Melino Muhammad*, **Zechen Li***, Flora Salim, Ahmed A. Metwally. *CGM-JEPA: Learning Consistent Continuous Glucose Monitor Representations via Predictive Self-Supervised Pretraining*. Preprint, 2026.
- **Zechen Li**, Baiyu Chen, Hao Xue, Flora D. Salim. *ZARA: Training-Free Motion Time-Series Reasoning via Evidence-Grounded LLM Agents*. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL Oral), 2026.
- **Zechen Li**, Shohreh Deldari, Linyao Chen, Hao Xue, Flora D. Salim. *SensorLLM: Aligning Large Language Models with Motion Sensors for Human Activity Recognition*. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2025.
- **Zechen Li**, Anders Søgaard. *QLEVR: A Diagnostic Dataset for Quantificational Language and Elementary Visual Reasoning*. Findings of the North American Chapter of the Association for Computational Linguistics (NAACL Findings), 2022.
- Meng Zhou*, **Zechen Li***, Pengtao Xie. *Self-supervised Regularization for Text Classification*. Transactions of the Association for Computational Linguistics (TACL), 2021. (Oral Presentation at NAACL 2021)
- Baiyu Chen, **Zechen Li**, Wilson Wongso, Lihuan Li, Xiachong Lin, Hao Xue, Benjamin Tag, Flora Salim. *AnyMo: Geometry-Aware Setup-Agnostic Modeling of Human Motion in the Wild*. Preprint, 2026.
- Ruiyi Yang, **Zechen Li**, Hao Xue, Imran Razzak, Flora D. Salim. *MAGE: Multi-Agent Self-Evolution with Co-Evolutionary Knowledge Graphs*. Preprint, 2026.
- Baiyu Chen, Wilson Wongso, **Zechen Li**, Yonchanok Khaokaew, Hao Xue, Flora Salim. *COMODO: Cross-modal Video-to-IMU Distillation for Efficient Egocentric Human Activity Recognition*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp), 2026.
- Meng Zhou, **Zechen Li**, et al. *On the Generation of Medical Dialogs for COVID-19*. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL Short), 2021.

Experience

Google, Student Researcher

- Building foundation models for continuous glucose monitoring.
- Connecting sensor foundation models with language models.

Mountain View, CA
Feb 2026 – Present

University of New South Wales, Casual Academic & Research Associate

- Taught tutorials for multiple courses:
 - COMP6713 Natural Language Processing
 - COMP9444 Neural Networks and Deep Learning
 - COMP9517 Computer Vision
 - DATA9001 Fundamentals of Data Science
- Conduct research on leveraging LLMs for wearable sensor time-series, focusing on human activity recognition and reasoning over motion sensor data.
- Mentored undergraduate students, co-supervising "Taste of Research" projects with Prof. Flora Salim.

Sydney, Australia
Aug 2022 – Dec 2025

Singapore University of Technology and Design, Visiting Research Fellow

Supervisor: Prof. Wei LU

- Conducted research on integrating human mobility data and time-series signals with language models.

Singapore
Mar 2023 – Jun 2023

University of Copenhagen, Research Assistant (part-time)

Supervisor: Prof. Anders Søgaard

- Developed QLEVR, a large-scale synthetic dataset (100k images, 1M questions) for visual reasoning with quantificational language.
- Designed scene graphs and 3D image generation pipeline using Blender to support complex linguistic and spatial reasoning tasks.

Copenhagen, Denmark
May 2021 – May 2022

University of California, San Diego, Research Assistant (part-time)

Supervisor: Prof. Pengtao Xie

- Proposed and evaluated a Self-Supervised Regularization (SSL-Reg) framework for text classification using a Y-structured joint model with shared parameters.
- Developed CovidDialog (English & Chinese), one of the first large-scale medical dialog datasets for COVID-19, and explored dialog generation with self-supervised regularization.
- Demonstrated that SSL-Reg alleviates overfitting and improves generalization, especially on low-resource tasks.

San Diego, CA
May 2020 – May 2021

Honors & Awards

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| • UNSW Founders New Wave Cohort 15 Showcase — Faculty of Engineering Award | 2025 |
| • UNSW Higher Degree by Research (HDR) Scholarship | 2023 – Present |
| • The Dean's Seattle Leaders & Scholars Award, Northeastern University | 2019 & 2020 |
| • Scholarship of Academic Excellence (Top 13%), Southern Medical University | 2016 & 2018 |
| • Excellent Student Leader (Top 30%), Southern Medical University | 2016 & 2018 |
| • 3rd Prize, South China Division, WeChat Mini Program Development Competition | 2018 |
| • Outstanding Student (Top 10%), Southern Medical University | 2016 |

Skills

- **Languages:** Chinese (native); English (proficient)
- **Programming:** Python, PyTorch, Java, R, C/C++, JavaScript, SQL